

Theme 6 - Vector algebra

Note Title

2012/05/20

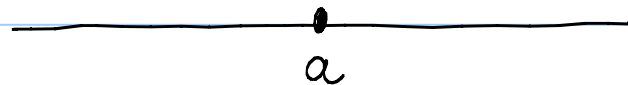
Unit 6.1 Three-dimensional coordinate systems.

Section 12.1 p 786 - 791

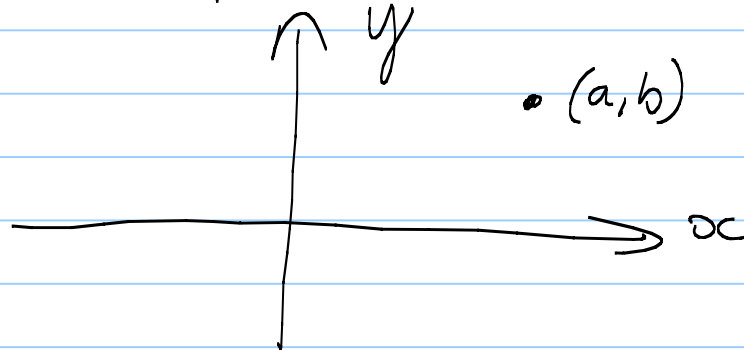
Notation $\mathbb{R}^n$
n-dimensional space

Examples

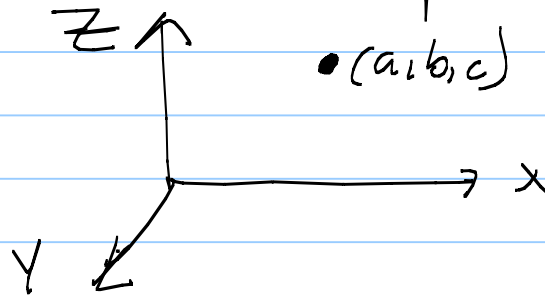
- $\mathbb{R}^1$ - one-dimensional space
- a line - with point a



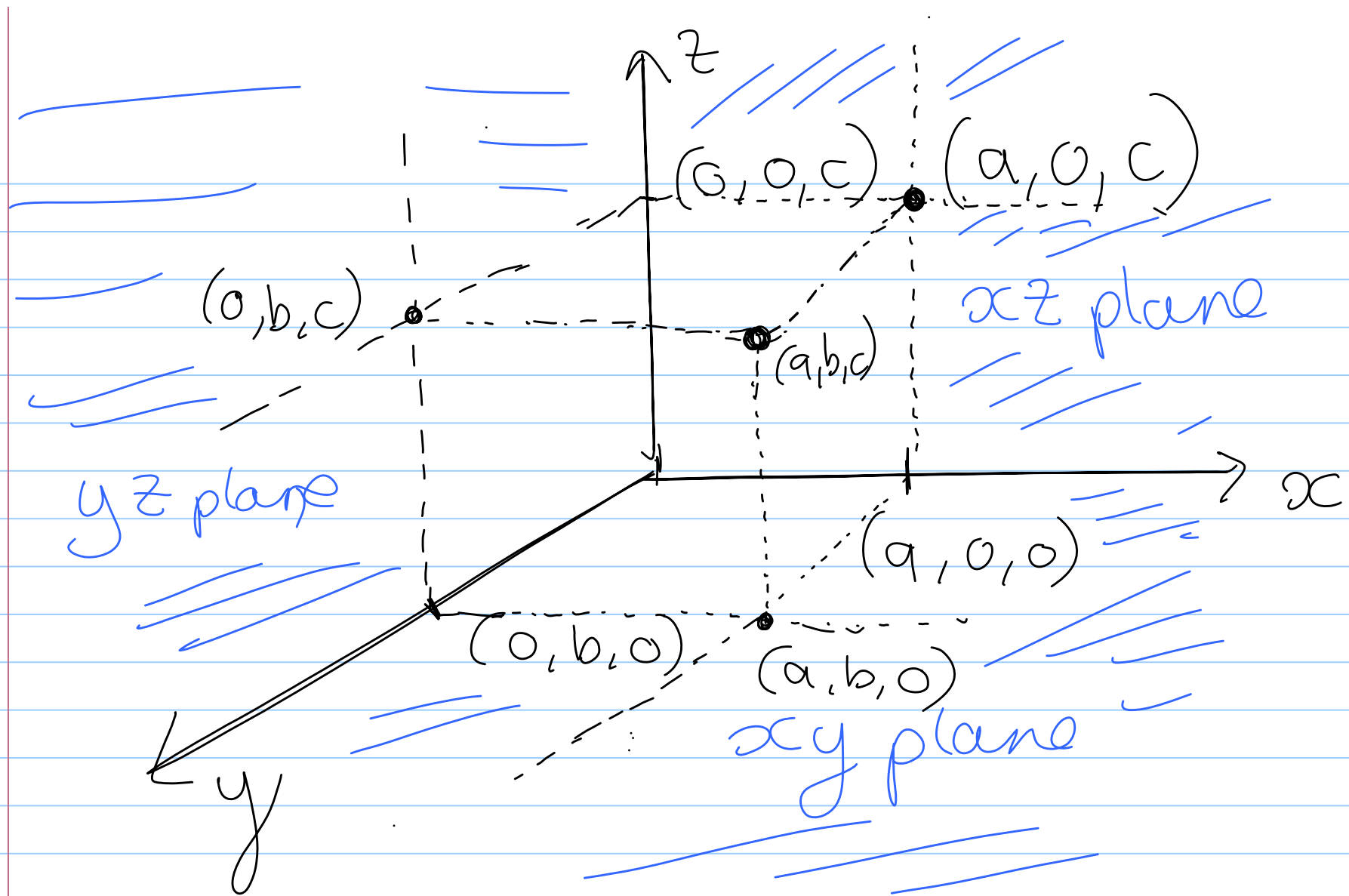
- $\mathbb{R}^2$ - two dimensional space
 - a plane, with point (a,b)



- $\mathbb{R}^3$ - three dimensional space
 - space, with point (a,b,c)

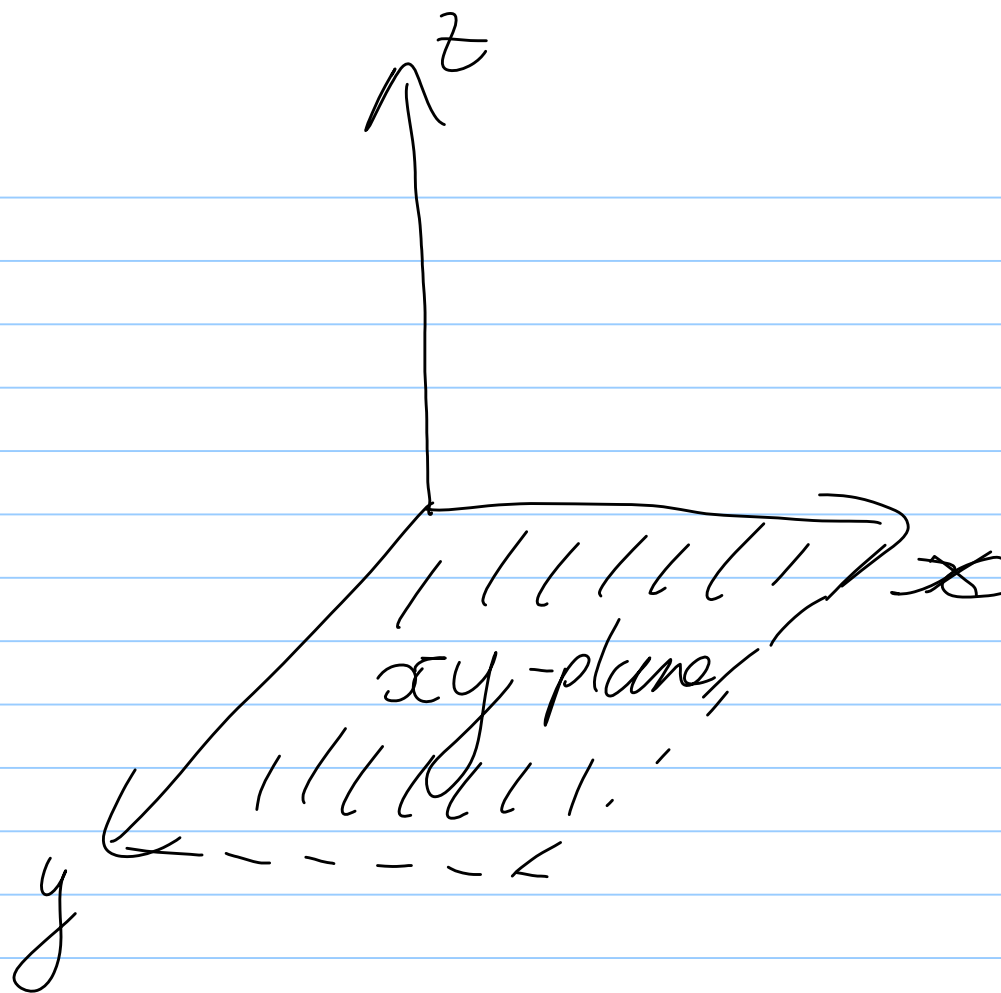
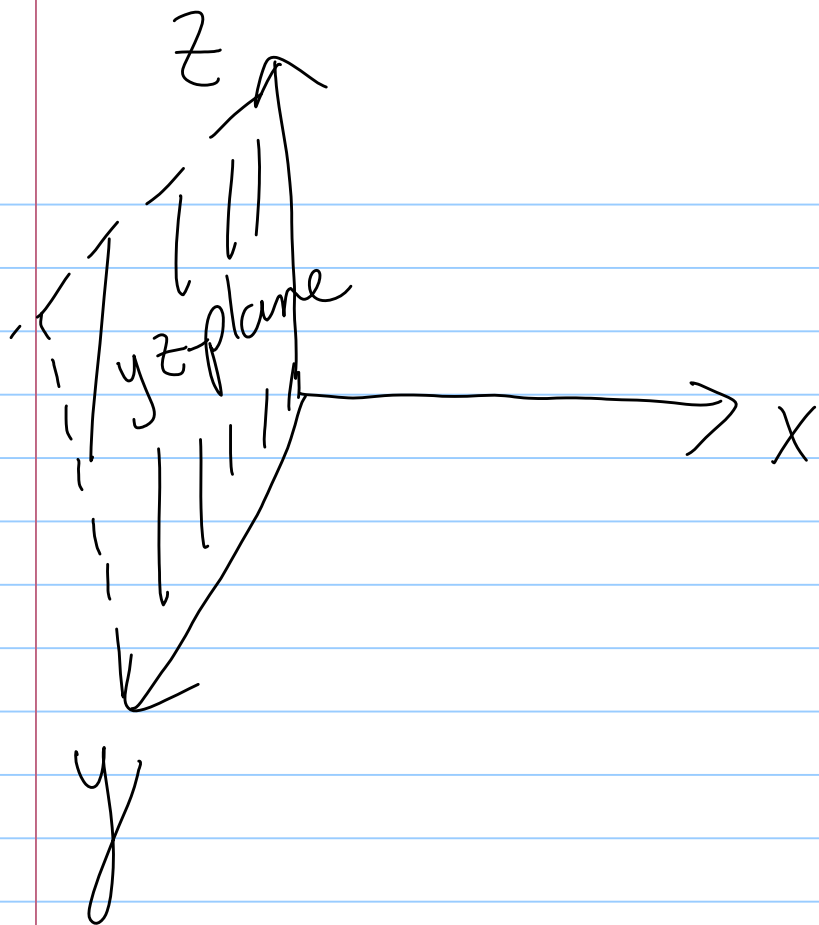


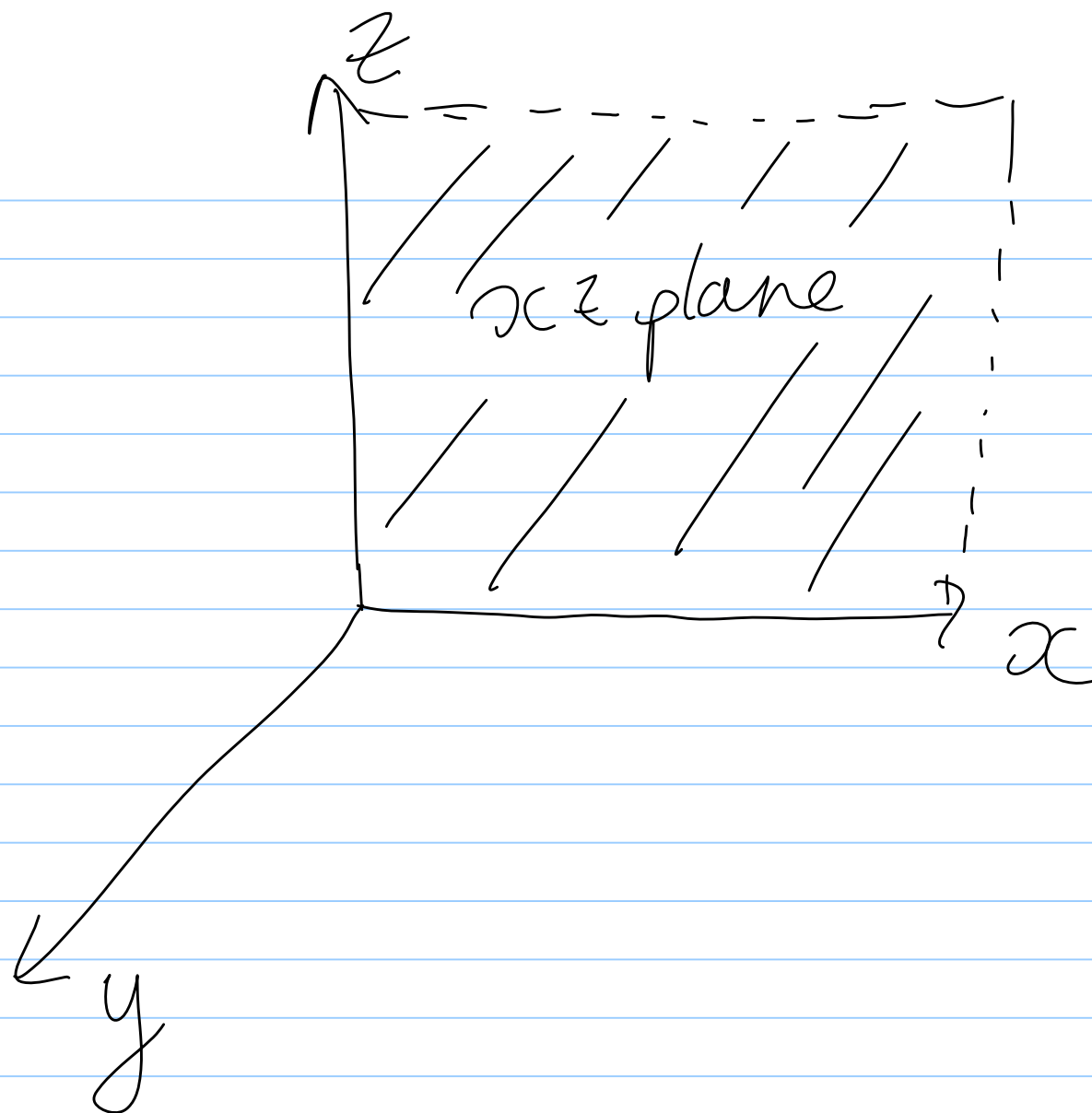
Example Sketch a cube, indicate planes, and all coordinates



Example : Sketch

- yz -plane with equation $x=0$
- oxy -plane with equation $z=0$
- oxz -plane with equation $y=0$.





Distance Formula in Three Dimensions

The distance between 2 points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ is

$$|PQ| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

Example

① Find the distance between

$A(1, 2, -3)$ and $B(4, 0, -3)$

② Find distance between

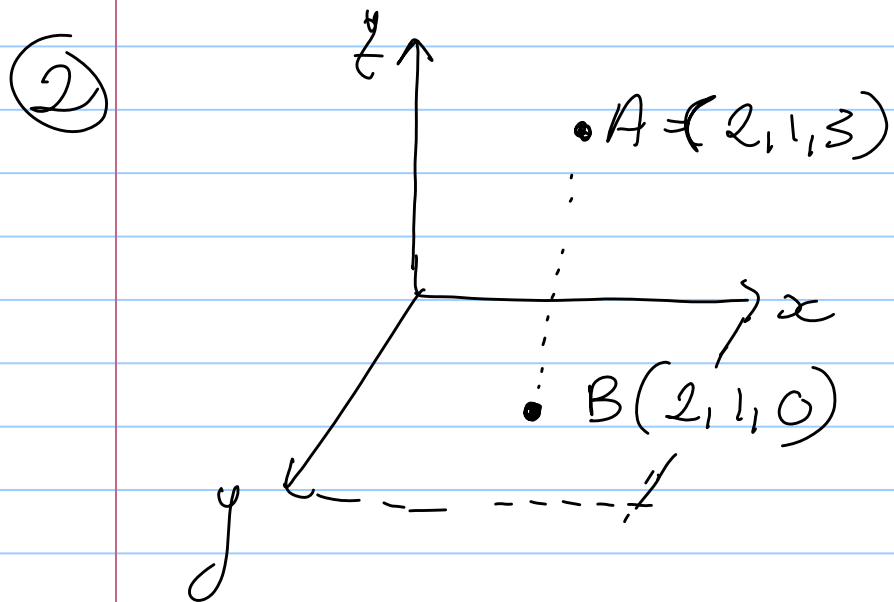
$A(2, 1, 3)$ and the xy -plane

③ Find the distance between

$A(3, 2, 4)$ and the y -axis.

Solutions

① $|AB| = \sqrt{3^2 + (-2)^2 + 0^2} = \sqrt{13}$



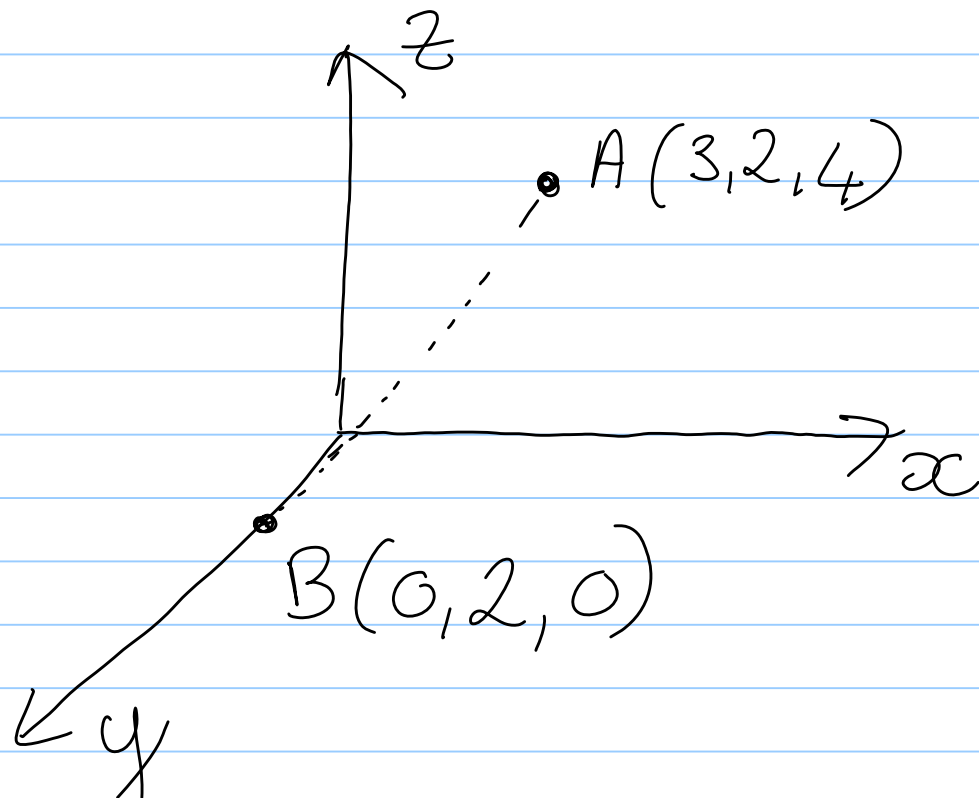
The projection
of A on the
xy plane is
 $B(2, 1, 0)$

∴ The distance is

$$|AB| = \sqrt{(2-2)^2 + (1-1)^2 + (3-0)^2}$$

$$= 3$$

3)



The projection
of A on
the y-axis
is B(0, 2, 0)

The distance is

$$|AB| = \sqrt{(3-0)^2 + (2-2)^2 + (4-0)^2}$$

$$= \sqrt{25}$$

$$= 5$$

Equation of a circle with center
 (a, b) and radius r is

$$(x-a)^2 + (y-b)^2 = r^2.$$

Equation of a sphere with a center
 (a, b, c) and radius r is

$$(x-a)^2 + (y-b)^2 + (z-c)^2 = r^2$$

Example Find the center and radius of a sphere

$$x^2 + y^2 + z^2 + 2x - 6z + 8 = 0$$

Solution

$$x^2 + 2x + y^2 + z^2 - 6z + 8 = 0$$

$$\Rightarrow x^2 + 2x + 1 - 1 + y^2 + z^2 - 6z + 9 - 9 + 8 = 0$$

$$\Rightarrow (x+1)^2 - 1 + y^2 + (z-3)^2 - 9 + 8 = 0$$

$$\Rightarrow (x+1)^2 + y^2 + (z-3)^2 = 2$$

$$\Rightarrow (x+1)^2 + (y-0)^2 + (z-3)^2 = (\sqrt{2})^2$$

The center is $(-1, 0, 3)$
and radius is $\sqrt{2}$

Example Sketch $y=3$ in $\mathbb{R}^2$ and $\mathbb{R}^3$

